

## UTTI, Finland

WMO: 029660

Lat: 60.891N

Lon: 26.941E

Elev: 339

StdP: 14.52

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |       |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|-------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |       | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB  | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)   | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -11.1      | -5.4 | -15.7                         | 2.4 | -10.8 | -9.4 | 3.4 | -5.1 | 20.9                  | 25.8 | 19.0 | 25.5 | 5.2                   | 340  | 0.630 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 16.1                   | 81.3            | 64.8 | 78.2 | 63.3 | 75.0 | 61.9 | 67.6                | 77.0 | 65.7 | 74.3 | 63.8 | 71.7 | 9.1                  | 200  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 64.5                            | 92.0 | 70.8 | 62.6 | 85.8 | 68.9 | 60.7 | 80.1 | 67.1 | 32.3          | 76.9 | 30.7 | 74.3 | 29.2 | 71.8 | 74.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 18.5          | 16.4 | 14.8 | DB                         | -15.1 | 85.0               | 7.5 | 3.7                                                | -20.5 | 87.7       | -24.9 | 89.8       | -29.0 | 91.9       | -34.5 | 94.6 | (4) |
| (5)               |      |      | WB                         | -15.0 | 70.4               | 7.7 | 2.4                                                | -20.5 | 72.0       | -25.0 | 73.4       | -29.3 | 74.7       | -34.8 | 76.5 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 40.8        | 19.7  | 19.7  | 26.8  | 39.0  | 50.1  | 58.6  | 64.1  | 60.9  | 51.6  | 40.7  | 32.0  | 25.1  |  |  |
|                                                                        | DBStd       | 17.58       | 12.03 | 11.95 | 8.13  | 6.90  | 7.81  | 6.33  | 5.91  | 5.40  | 6.03  | 7.04  | 8.12  | 10.90 |  |  |
|                                                                        | HDD50       | 4609        | 940   | 849   | 718   | 339   | 95    | 5     | 0     | 1     | 55    | 295   | 541   | 771   |  |  |
|                                                                        | HDD65       | 8934        | 1405  | 1269  | 1183  | 781   | 463   | 210   | 90    | 150   | 403   | 754   | 990   | 1236  |  |  |
|                                                                        | CDD50       | 1255        | 0     | 0     | 0     | 8     | 99    | 264   | 437   | 339   | 102   | 6     | 0     | 0     |  |  |
|                                                                        | CDD65       | 107         | 0     | 0     | 0     | 0     | 3     | 19    | 62    | 23    | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 878         | 0     | 0     | 0     | 0     | 42    | 175   | 495   | 166   | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH80       | 153         | 0     | 0     | 0     | 0     | 5     | 22    | 107   | 19    | 0     | 0     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 7.3         | 7.3   | 7.3   | 7.5   | 7.2   | 7.5   | 7.5   | 7.0   | 6.7   | 7.0   | 7.3   | 7.7   | 8.0   |  |  |
|                                                                        | PrecAvg     | 25.6        | 2.0   | 1.6   | 1.5   | 1.2   | 1.4   | 2.4   | 2.7   | 2.8   | 2.3   | 2.7   | 2.6   | 2.2   |  |  |
| (15) Precipitation                                                     | PrecMax     | 33.7        | 3.8   | 3.8   | 3.1   | 2.2   | 3.3   | 4.2   | 5.3   | 5.9   | 5.1   | 5.7   | 5.6   | 3.4   |  |  |
|                                                                        | PrecMin     | 19.3        | 0.4   | 0.2   | 0.2   | 0.4   | 0.4   | 0.6   | 0.3   | 0.2   | 0.7   | 0.6   | 0.3   | 0.6   |  |  |
|                                                                        | PrecStd     | 3.5         | 0.9   | 0.9   | 0.7   | 0.5   | 0.9   | 1.0   | 1.3   | 1.4   | 1.4   | 1.3   | 1.2   | 0.8   |  |  |
|                                                                        |             |             |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 39.0  | 40.4  | 49.6  | 67.2  | 79.0  | 82.7  | 87.0  | 82.4  | 71.0  | 57.4  | 48.8  | 45.5  |  |  |
|                                                                        |             | MCWB        | 38.4  | 36.4  | 41.1  | 52.3  | 59.8  | 63.9  | 67.2  | 65.0  | 60.3  | 53.3  | 47.7  | 44.4  |  |  |
|                                                                        | 2%          | DB          | 37.0  | 36.9  | 43.1  | 60.8  | 73.2  | 78.8  | 82.9  | 78.5  | 66.7  | 54.6  | 46.5  | 41.9  |  |  |
|                                                                        |             | MCWB        | 36.4  | 35.4  | 37.1  | 48.5  | 57.6  | 62.0  | 66.4  | 63.7  | 58.3  | 52.2  | 45.8  | 41.2  |  |  |
|                                                                        | 5%          | DB          | 35.4  | 35.1  | 39.3  | 55.8  | 68.8  | 75.2  | 79.4  | 75.1  | 64.0  | 52.4  | 44.6  | 39.5  |  |  |
|                                                                        |             | MCWB        | 35.0  | 34.3  | 34.9  | 45.9  | 54.6  | 61.2  | 64.8  | 62.4  | 56.9  | 50.6  | 44.1  | 38.8  |  |  |
|                                                                        | 10%         | DB          | 33.6  | 33.5  | 36.5  | 51.2  | 64.6  | 71.4  | 75.8  | 71.6  | 61.0  | 50.4  | 42.7  | 37.4  |  |  |
|                                                                        |             | MCWB        | 33.2  | 32.9  | 33.8  | 43.0  | 52.6  | 58.8  | 63.1  | 61.2  | 56.1  | 49.0  | 42.1  | 36.8  |  |  |
|                                                                        | 0.4%        | WB          | 38.4  | 37.4  | 41.7  | 55.1  | 63.5  | 67.9  | 70.7  | 69.0  | 62.2  | 55.1  | 48.1  | 44.3  |  |  |
|                                                                        |             | MCDB        | 38.5  | 38.4  | 47.7  | 63.5  | 73.6  | 77.8  | 80.3  | 77.8  | 67.2  | 56.3  | 48.7  | 44.8  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB          | 36.1  | 35.6  | 38.0  | 50.5  | 59.5  | 65.3  | 68.5  | 65.9  | 60.2  | 53.1  | 45.8  | 41.3  |  |  |
|                                                                        |             | MCDB        | 36.4  | 36.5  | 41.6  | 59.0  | 68.0  | 74.7  | 78.5  | 73.3  | 64.4  | 54.2  | 46.3  | 41.7  |  |  |
|                                                                        | 5%          | WB          | 34.8  | 34.2  | 35.9  | 47.0  | 56.6  | 62.8  | 66.9  | 64.1  | 58.6  | 51.3  | 44.0  | 38.9  |  |  |
|                                                                        |             | MCDB        | 35.1  | 34.9  | 38.1  | 54.1  | 66.3  | 71.4  | 76.3  | 71.5  | 62.5  | 52.3  | 44.4  | 39.4  |  |  |
|                                                                        | 10%         | WB          | 33.1  | 32.8  | 34.2  | 43.9  | 53.7  | 60.3  | 65.0  | 62.5  | 56.8  | 49.3  | 42.1  | 36.9  |  |  |
|                                                                        |             | MCDB        | 33.3  | 33.3  | 35.8  | 49.9  | 63.1  | 69.0  | 73.4  | 69.3  | 60.2  | 50.3  | 42.4  | 37.3  |  |  |
|                                                                        | 5% DB       | MDBR        | 9.2   | 10.3  | 12.9  | 15.6  | 18.1  | 16.2  | 16.1  | 15.5  | 13.3  | 9.4   | 7.2   | 8.2   |  |  |
|                                                                        |             | MCDBR       | 6.8   | 8.1   | 15.6  | 23.5  | 24.1  | 22.2  | 20.8  | 20.6  | 16.9  | 10.8  | 8.0   | 7.0   |  |  |
|                                                                        |             | MCWBR       | 6.9   | 7.4   | 10.9  | 13.6  | 11.9  | 10.3  | 8.6   | 8.6   | 9.6   | 8.3   | 7.8   | 6.9   |  |  |
|                                                                        |             | MCWBR       | 6.8   | 7.4   | 12.5  | 21.3  | 21.0  | 19.1  | 18.5  | 17.2  | 14.4  | 9.9   | 7.7   | 6.3   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.262 | 0.302 | 0.326 | 0.364 | 0.361 | 0.357 | 0.380 | 0.368 | 0.344 | 0.325 | 0.287 | 0.373 |  |  |
|                                                                        |             | taud        | 2.106 | 2.167 | 2.236 | 2.335 | 2.418 | 2.458 | 2.414 | 2.437 | 2.496 | 2.403 | 2.187 | 2.612 |  |  |
|                                                                        | Ebn at Noon |             | 147   | 212   | 250   | 263   | 274   | 277   | 266   | 260   | 247   | 207   | 136   | 100   |  |  |
|                                                                        |             | Edh at Noon | 13    | 23    | 30    | 33    | 32    | 32    | 32    | 29    | 23    | 18    | 12    | 8     |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 77    | 268   | 740   | 1208  | 1641  | 1716  | 1705  | 1303  | 765   | 318   | 85    | 40    |  |  |
|                                                                        | RadStd      |             | 16    | 58    | 92    | 129   | 181   | 109   | 161   | 142   | 89    | 49    | 15    | 5     |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |  | +0.83 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (52 neighbors) |  | +0.90 | N/A     | N/A    | N/A     | N/A   | N/A   | -290        | -308  | N/A   | N/A   |

Nomenclature: See separate page